



THE NEW STACK

AI for the Enterprise

**The Playbook for Developing
and Scaling Your AI Strategy**

Jennifer Riggins

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The New Stack

AI for the Enterprise: The Playbook for Developing and Scaling Your AI Strategy

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CHAPTER 01

Standing at the AI Inflection Point: Where AI Promise Meets Enterprise Practice

There's no debate that enterprises are rapidly adopting AI technology; Couchbase's FY 2026 CIO AI Survey pegged [AI investment growth at 52% year over year](#). The vast majority (87%) of enterprises surveyed declared that their organizations must embrace AI in the next six months or fall irrevocably behind.

But too much pressure to adopt AI without a top-down strategy puts your organization's long-term reliability and security at risk — alongside your employees' trust. It's time for executive-level leadership to enact an internal AI strategy that aims to empower existing staff — not to replace them or just splatter AI everywhere.

This guide shares the lessons from what's working at multinational organizations — and what's not. It's both a snapshot in time at the end of 2025, and a look ahead to prepare for the next couple years of inevitable change. By setting your organization up for success today with clear communication, leadership buy-in, and skills development — along with a keen focus on data protection and security built into scalable architecture — your company will be ready for the demands of generative AI today and the agentic AI-empowered work of tomorrow.

This book lays out how AI can address your organization's biggest challenges while also chipping away at the toil and boredom that distract from delivering the best work quickly. Its focus is on incremental, measurable improvement and the architectural and process flexibility necessary to pivot quickly in response to continuous change.

This playbook also aims to help you scale your AI initiatives using the best practices already delivering in your organization: hybrid and multicloud, open source, platform engineering, progressive delivery, DevOps, and breaking down silos. It will show you how to set up an end-to-end workflow management system to steer your MLOps program. And it will give you the frameworks to measure return on this massive AI investment.

Throughout, you will find enterprise-grade AI use cases that have already been proven — for both software development teams and more broadly across organizations. This grounds your AI strategy in what's working while

inspiring a culture of continuous experimentation and learning that will eventually see AI solve some of your toughest problems.

Artificial intelligence is no longer a fad. AI must be a measurable strategy. I hope you will learn from this book and the leaders interviewed throughout. And I hope you will share your own AI growth journey. Only by sharing questions and concerns, AI wins and losses, will we start to see collective benefit from AI.

The Conflicting State of AI in 2025

As with anything at such scale and under such pressure, there are challenges and opportunities.

The [Atlassian 2025 “State of Developer Experience Report”](#) found that 99% of developers reported time savings with AI tools, with 68% claiming to save more than 10 hours per week because of AI. [According to a recent McKinsey survey](#), engineers say they are happier, more able to focus on satisfying and meaningful work, and more able to achieve “flow state” with generative AI (GenAI).

While perception of productivity does have a positive impact on reality, developers’ overall enjoyment of AI doesn’t mean they are as productive as they think — at least not yet.

AI research nonprofit Model Evaluation and Threat Research (METR) found that, despite predicting they would be 24% faster, [developers were 19% slower](#) when using AI to complete tasks. Respondents observed that AI-generated code resulted in “more to fix up,” in terms of code conventions

and style, and they had to “do a lot of simplifying its code.”

This result isn’t an outlier. Harness’s [“State of Software Delivery 2025”](#) found that 67% of developers spend more time debugging AI-generated code, while 68% spend more time resolving security vulnerabilities.

And this goes beyond the individual level. The [2024 DORA report](#) found that, organization-wide, a 25% increase in AI adoption led to a 7.2% decrease in delivery stability and a 1.5% decrease in delivery throughput. And MIT NANDA found that [most AI pilots fail](#), pointing to a lack of shared, contextual learning and continued experimentation inside data silos.

Distance Between Leaders and Developers

Despite [top-down buy-in across executive boards](#), there's still a disconnect between leadership and engineers over AI.

According to LeadDev's "[2025 Engineering Leadership Report](#)," 60% of engineering leaders don't believe AI has significantly boosted their team's productivity.

Top-down AI mandates can't come at the cost of ignoring developers. In Atlassian's DevEx survey, 63% of developers say leadership doesn't understand their challenges — up from 44% the previous year.

This disconnect extends past tech teams too. Without well-communicated AI strategies, [half of all knowledge workers are using shadow AI](#) — AI tools they've adopted (and even paid for) outside the consent and firewalls of their employers, and 46% wouldn't give up their AI use — even if it were banned, found Software AG.

Clearly your employees are finding value from AI, which cannot be ignored. But when quality, speed, and security are at risk, engineering leaders must measure AI's impact on the whole software development life cycle and their employees' life cycle to better understand how to close that gap.

While teams should be encouraged to experiment with AI, it can no longer be done in isolation. Top-down leadership is necessary to enact an AI strategy that helps both technical and nontechnical teams safely increase efficiency.

CHAPTER 02

Set Your Organization Up for Success

AI demands massive investment. While many early AI use cases have come from the bottom up, an enterprise-grade AI strategy must be top down. And, while it's delivered via technology, the people and processes are just as key.

Only by creating a top-down AI strategy that's well communicated to all stakeholders, inside and outside your organization, can you enable the learning, training, and sharing that will allow you to transform into an AI organization.

A New Seat at the Table: Chief AI Officer

An AI strategy can't just be bolted onto an existing executive role at any larger organization. The CEO has to run the business, and every other C-level role must remain focused on their areas of expertise — be it finance, people, technology, or operations — in order to bring AI-driven productivity to every team.

Along with everything else, AI is disrupting the corporate hierarchy. AI is so big that it demands a new seat at the boardroom table: [the chief AI officer \(CAIO\)](#).

The CAIO's job isn't about driving pie-in-the-sky innovation. It's about laying down the guardrails and gates that allow organizations to develop and execute an AI strategy.

"The CAIO isn't building the next rocket; they're making sure it doesn't crash on launch," [writes tech recruitment firm Pltfm](#). "Their job? To manage AI-related risks (such as ethics), as well as adopt, scale, and integrate systems responsibly."

What To Look For in a CIAO

- AI domain expertise
- AI-related ethics, security, privacy, and data protection awareness
- Deep product and customer knowledge
- Collaborative spirit, ability to foster partnerships across business and IT
- Ability to embrace experimentation
- Strong communications skills

While it can be tempting to rebrand your CTO as “chief of technology and AI,” that will likely land your organization in the same chasm that’s divided technology and business from the beginning. While technology is delivering this innovation, you need someone with a keen eye for data protection, privacy, and security, as well as deep product knowledge, domain experience, and understanding of your customers. You also need someone who can bring together representatives from each department to find ways that can help increase productivity within responsibilities.

“Having a CAIO means viewing AI not just as a tool, but as a core component of your strategic vision, positioning your company as a leader in the field,” said Barry Guiney, COO of Odgers Berndtson Ireland executive talent firm, in [Silicon Republic](#).

“They create an environment where AI specialists and data engineers can thrive and innovate. They foster key partnerships that are crucial for

leveraging the latest AI advancements, and with a dedicated leader, these companies can navigate the complexities of AI integration more effectively.”

This unified role not only heads the AI strategy and communication for your organization, but also facilitates a program of sandboxed experiments that enables shared learning, reusability, and cross-organizational AI-driven productivity.

“Strategic focus must be on AI as an enabler, an augments of work, and automator of the tedious.

Corporate AI Communications: The Good, the Bad, and the Ugly

For years, we've been bombarded with headlines of how AI is going to steal our jobs. In 2026, this messaging will harm your organization.

Strategic focus must be on AI as an enabler, an augmenter of work, and automator of the tedious. This way, at all stages, in all roles, people can focus on solving more complex problems. Make the priorities of your AI strategy clear with top-down messaging.

"The best communication is from the top," said Red Hat Senior Product Marketing Manager Marty Wesley. "The message isn't, 'Your job is going away.' It's, 'Use this [AI] to find out how you can be more productive.'"

Wesley's team, which now works with 200 top enterprise clients, is exploring ways to augment its work with AI to serve 350 clients with the same quality. For customer marketing, that includes using AI to help understand nuances of particular industries to identify their top concerns. What would have taken

weeks to research now takes an afternoon, he reflected, enabling a more personalized touch at more accounts.

"It's not about job loss. It's about expanding our work funnel," he remarked. "That's the best communication method: Talk about how it's going to augment or expand opportunities."

One thing's for sure, we have plenty of lessons on how to communicate top-down AI mandates.

Implement in Phases

For example, Box CEO Aaron Levie laid down an AI strategy in two phases. First came the [internal employee messaging on building an AI-first company](#):

- Use AI to eliminate drudgery and move faster across the business.
- Encourage teams to use AI to automate more and save money, but reinvest those savings.

- Foster constant experimentation internally to find the best use cases for AI.
- Upskill every employee to be AI-first over time with more education and awareness.
- Maintain strong governance and security practices, with human oversight in most areas.

In phase two, Box released its [customer-facing AI principles](#) and commitments for using AI in its document management products:

- Full customer control of AI usage.
- No training models using customer content without explicit approval.
- Explanations of AI output.
- Strict adherence to permissions.
- Data security.
- Transparency.
- Protection of user and enterprise data.
- Trustworthy AI models.

Convey Empathy

While AI can and should be a top-down initiative, success comes from listening to your employees.

For example, when Canva's employees said they needed time and space to "get curious" about AI, the company [paused all business](#) for a week to allow its 5,000 employees to focus on "building, learning, and experimenting" with AI.

Executives emphasized the design tool's guidelines around responsible AI usage and outlined three focus areas:

- Internal workflows to boost productivity.
- Enhance existing features or invent something entirely new for the product.
- Inclusive, accessible solutions for the community.

Over five days, this top-down initiative spurred 330 hackathon ideas and a collective 25,000 hours of learning.

Be Mindful of Your Messaging — Internal and External

No matter what your AI strategy is, people will be talking about it as soon as you announce it. So be mindful of what you say.

In April 2025, [Shopify CEO Tobias Lütke announced that AI is mandatory](#) for all employees and nonuse of AI must be justified. Furthermore, no one working at the e-commerce giant is allowed to hire a new human without first justifying that AI cannot do that job.

Around the same time, [Micha Kaufman, CEO of freelance marketplace Fiverr, wrote](#): “So here is the unpleasant truth: AI is coming for your jobs. Heck, it’s coming for my job too. This is a wake-up call.”

He then specified some of the most ubiquitous roles within any organization — legal, customer support, finance, sales, and programming — that AI may

absorb, and said becoming an “exceptional talent in your field” is required to maintain your career.

Some of Kaufman’s AI adoption advice is prudent:

- Try multiple AI solutions and figure out what gives you superpowers, which he defined as the “ability to generate more outcomes per unit of time with better quality per delivery.”
- Find the most knowledgeable people on your team to learn from.
- Understand the company strategy well and contribute to achieve its goals.

Indeed, the famous tech–business chasm is no longer an excuse. All teams must be focused on business goals, and the purpose of using AI must be to help them achieve goals faster.

Consider Your External Stakeholders

Remember that communicating your AI strategy isn't just for your internal audience — especially when you deploy AI into your products and services.

For example, Duolingo announced [148 new language courses generated with AI](#), doubling the number it offers. To continue moving “with urgency,” CEO [Luis von Ahn said](#) the company would no longer use contractors or increase headcount for work that can be automated with AI, and AI usage would be part of performance review evaluations.

After von Ahn was publicly criticized, he clarified that no full-time staff layoffs were planned. Despite the online criticism, the publicly traded company's stock soared in response to his AI-first announcement, showing that at least investors accept AI as the secret to success.

In the end, no matter how “all-in on AI” your org is, how you communicate your AI strategy sets you up for success or failure.

Create an AI Policy

Most organizations needed an AI usage policy yesterday. Yet only about half have them.

“How many products have had AI shoved in there while completely ignoring the primary pain points of their products and their platforms?” asked Hannah Foxwell, founder of AI for the Rest of Us. “Your AI strategy needs to be grounded in the business today, not in some fictitious version of the business tomorrow.”

To develop your AI policy, establish a cross-functional, cross-organizational AI enablement team — perhaps seated in the new CAIO office.

The [International Organization for Standardization's AI standards](#) are a good place to start considering your AI policy:

- **ISO/IEC 42001 – AI governance and risk management:**
Establishes accountability and oversight, setting up risk management protocols to identify, access, and mitigate potential risks and impacts.

- **ISO 27001 – Information security:** Recommends tooling that vets across people, processes, and technology. This includes a privacy management extension.
- **ISO 37301 – Compliance and ethical culture:** Defines requirements for compliance management systems, as well as guidelines to nurture a culture of integrity, honesty, and fairness in AI usage.

[The European Union's AI Act](#) offers another good way to classify risk around AI tooling:

- **Unacceptable AI:** Includes scraping the internet or security cameras to build facial recognition databases.
- **High-risk AI:** Includes any AI for critical infrastructure and utilities, education, and employment management.

- **General-purpose AI models:** For AI systems that are transparent by default, which allows users to make informed decisions.

This categorization may vary for different parts of your business, depending on their levels of risk acceptance or avoidance.

It's also wise to give everyone in your organization a refresher in local and international data privacy regulations, like the EU's GDPR and the California Privacy Rights Act (CPRA). GenAI tools' ease of use and adoption have already led to [inadvertent data leakage](#) — you don't want to be next.

Communicate Your AI Policy

Once your AI enablement team agrees on an AI usage policy, make sure you communicate it with all stakeholders inside and outside the organization. Grammarly has been using AI in its product for the last 15 years, and its

[User Trust Center](#) has become an industry standard for clear AI communication. It breaks the risks down into four areas:

- Privacy
- Security
- Compliance
- Responsible AI

“Managers and leaders are going through this learning journey alongside their teams, so you can’t look to your manager for concrete advice,” Foxwell said. “A centralized enablement team or external training partner is a good way to go if you want to get consistent messaging.”

Don’t forget that your AI strategy must also factor the Software as a Service (SaaS) products you are already using. If they’ve injected some sort of AI into the product, has your company vetted it for compliance?

Since these risks and use cases vary by domain, you can’t have one policy to rule them all. Consider some overarching data and security rules, but then allow subject matter and domain experts to weigh in on department-specific aspects. And ground any policies in examples based by domain or department.

As you create this AI usage policy, just remember to “translate” it from legalese to lay terms so that all employees understand. This is a good use case for generative AI — so long as you have a human reviewer in the loop.

“Since these risks and use cases vary by domain, you can’t have one policy to rule them all.”

CHAPTER 03

Take the Incremental Approach

Everything is changing so fast that speed to market may come by slowing down. Instead of taking the “AI full-speed-ahead” route, sustainable AI adoption and return on investment (ROI) hinge on setting expectations, running isolated tests, and showing value before rolling proven use cases out to broader staff.

“Leaders have a real tendency to want everything all at once,” Hannah Foxwell said. “Wise people are optimizing for fast feedback and learning, knowing that will get them more in the long run.”

Give people space to tackle their own problems. Teams know best the things holding them back, but they may not feel able to prioritize them. Schedule time away from regular feature delivery to focus on studying and experimenting with AI.

Two-Speed AI Investment

One big way to close this gap, Red Hat's Marty Wesley advised, is to follow the 80/20 rule. Spend 80% of your AI budget on established use cases — not just across the software development life cycle but across the full employee experience. Then, spend 20% of your time and money on making big bets: things that will unlock real ROI if AI can deliver on them. And if it can't yet, then at least you didn't bet the whole business.

AI has plenty of proven enterprise-grade use cases that can increase productivity in small ways, like documentation management, service ownership discovery, and anything that helps employees find internal documents and processes.

"They're the kind of projects that you know are going to work because lots of other organizations have been successful, so there's a known pattern," Wesley said. For example, he said, Red Hat's large customer support organization is using AI to pull up answers from existing documentation to improve response time.

Proven, incremental improvements like these add up. As Wesley put it, "If you stack up pennies, you suddenly start to get some real money. And these are pennies you know you'll get."

This probably isn't the huge transformative AI impact you're hoping for.

At the other end of the spectrum, Wesley said, is when senior leadership picks "the biggest, hardest, most challenging problem" your organization is facing, like migration to the cloud. "Get your strategy and your R&D people that are already looking out to the future. See if they can have a run at it," he advised.

This 20% AI investment could lead to a 10% improvement on a large, hard-to-solve problem.

"You're taking a risk because it may or may not work, but you'll probably make headway," he explained.

Case Study: Turkish Airlines' AI Strategy

[Turkish Airlines' AI strategy](#) is to embrace AI-assisted operations across all of its processes and services for its nearly 90,000 employees. Its strategy doesn't just focus on developer productivity, but takes a broader view to increase satisfaction and productivity for all employees through AI augmentation.

On the tech side, the airline sought an open source AI strategy in order to leverage a hybrid cloud environment while avoiding vendor lock-in.

Turkish Airlines kicked off its AI strategy with data. From Day 1 of its AI strategy, the airline cleansed all its data sources.

"It doesn't matter who you are. If you want to experiment with AI or you want to build an AI application with a few clicks of a button or self service, [you need] audited, controlled access to all of our major data sources," explained Joe Giordano, chief architect in the Red Hat field CTO organization.

"Platforms like Red Hat OpenShift AI allow us to keep data on premises while accelerating model development and enabling business units to leverage AI capabilities in a flexible way," said Serdar Gürbüz, general manager of Turkish Technology, a Turkish Airlines subsidiary.

[Red Hat OpenShift AI](#) is a platform for managing the life cycle of predictive and generative AI models at scale across hybrid cloud environments. It enables provisioning and autoscaling capabilities with automatic resource reallocation. This allows multiple data scientists to use the same datasets equally, while operations can focus on value-added work.

Through this centralized open source platform, Turkish Airlines was able to give development and data science teams more autonomy, spinning up custom development environments, automatic deployment for AI models,

pipelines, and custom monitoring and alerts — starting from the AI exploration stage all the way through to production.

Finally — perhaps the most important part of any enterprise transformation — Red Hat OpenShift AI fostered a measurable link between the technical and business AI strategy, boosting revenue while reducing operation costs.

The proof of AI success is usually in the use cases delivered that better serve an individual organization or industry. During the first half of 2025, Turkish Airlines was able to make several significant improvements across 60 live models with 40 more in development. These included:

- Improved accuracy of dynamic pricing models for airline ticket sales.
- Enhanced models for fraud protection.
- Customer service chatbots
- Aircraft assignment optimization, cutting fuel consumption.
- Improved ground plan prediction, block time prediction, and on-time performance prediction, enabling operations to better predict and act on delays.

The airline has already doubled its deployment speed and [achieved \\$10 million in savings](#) from AI-enhanced flight planning and fuel-loading processes alone.

"As Turkish Airlines continues its journey to becoming a fully data-driven organization, our focus is on embedding AI across every aspect of the business," Gürbüz said. By embracing hybrid cloud via Red Hat OpenShift AI, the airline is better able to "move AI beyond isolated use cases and into a scalable, organization-wide capability."

AI Use Cases (Beyond Code Generation)

If writing code was never the bottleneck and code generation is already widely adopted, then organizations need to look elsewhere for new AI use cases.

There will be cases of one-size-fits-all AI. SAP's most popular AI use case by far is scanning and processing receipts — something many employees do. But more likely, AI will solve for smaller pains, often specific to roles or functions. Keep things lean and simple by starting with early wins that could stably grow in the future. Then, once a use case is proven, you can adapt it for other teams.

While there are plenty of non-technology problems to solve, the first AI use cases tend to be incubated within tech teams, Foxwell said. It feels safer to fail in the tech division because the internal data used by tech teams may not have as much impact as finance, human resources, and customer data. AI could have greater impact in those areas, but they also have much higher risk.

Non-Coding AI Use Cases to Try

- Improve documentation quality and access
- Modernize legacy applications and manage technical debt
- Speed code review and quality assurance
- Automate repeatable processes
- Unlock insight in unstructured data

Documentation and Technical Debt

Developers are consistent in explaining [what most slows them down](#): insufficient, out-of-date documentation and technical debt. Neither of these are issues most devs want to fix themselves. Both also worsen with a rapid increase in code — and both are use cases where AI can shine because it is great at having conversations and explaining complexity.

In fact, the biggest AI win uncovered in the 2024 DORA report found that a 25% increase in AI adoption triggered a 7.5% increase in documentation quality. Take this use case company-wide by adding a chatbot overlay for internal information, ownership, and processes.

Rachel Laycock, CTO of Thoughtworks, finds that legacy modernization is still the top challenge facing most enterprises. “It’s not clear when we will get to a settled space of the top three or five tools and models people use,” she said.

Additionally, she said, people are most often building “greenfield” apps, which is easy compared to building features that have to be integrated into legacy applications. The market, she continued, is “focusing too much on efficiency of producing code, which isn’t actually the problem.”

Code Review

Another bottleneck that increases with development speed is code review, which leaves developers waiting longer for feedback and increases cognitive load.

“Previously, quality was derived from having human eyes at two steps — the developer who wrote it and the PR [pull request] reviewer,” Foxwell said. “Now the developer who ‘wrote’ it may not have read it. It may have been written by AI. It looks fine to ship, and the person reviewing it is faced with a mountain of code to review all at once.”

AI is also useful here, observed DORA Lead and Product Manager at Google Cloud Nathen Harvey. By having the right specifications, documentation, and guardrails in place, you can provide rapid, automated feedback to developers using AI as a first-round PR reviewer. Then AI can assign the PR to the peer with the availability and the knowledge to review it. This review also takes less time and is more constructive because this “human in the loop” can be confident that the PR adheres to established organizational standards.

Unlocking Insight in Data

Some organizations, like in financial services or governments, have 100 years’ worth of data that they can finally unlock with AI. However, that process may not be an easy one, especially within budgetary, regulatory, privacy, and security guidelines.

When thinking about AI use cases, Wesley remarked: “The easiest ones are in financial services, because there’s

just so much data, and they can use the AI to help collect a lot of that information and make decisions on it.” Consider the often tedious, in-person, days-long experience of opening a commercial account, which requires companies to submit all kinds of structured and unstructured data across different documents.

AI can read the existing forms and get a sense of the business, allowing the bank to move the whole process online in a much shorter time frame. And this isn’t justifying staff layoffs at these “accidental tech companies,” but instead restructuring for value by reassigning staff to other areas that have issues, problems, or just need more help, Wesley recommended.

Of course, setting up an account is a repeatable AI use case common to all businesses, from car insurance to university applications.

Another AI-driven use case is scanning unstructured data in images and videos more rapidly than humans could ever

imagine. For example, Boston Children’s Hospital and a transportation company are using Red Hat technology to [uncover patterns of disease](#) and [inspect railway tracks](#) using drones, respectively.

Boston Children’s Hospital says its AI radiology tools “are as good as the human radiologists, and they’re going to get better,” Wesley said.

“No organization can afford to adopt every tool out there, but they also can’t afford to miss out on the next AI productivity gain.

Safe Experimentation

If you think you had tool sprawl before, wait until you add in AI. No organization can afford to adopt every tool out there, but they also can’t afford to miss out on the next AI productivity gain.

Enterprises need a robust culture of experimentation supported by clear permissions, processes, and guidelines.

A good first step is encouraging teams to A/B test different products and offer feedback, like software development company Devexperts did when [testing and comparing code generator tools Copilot and Cursor](#), then share test methods and results.

At this speed of change, your chief AI office will likely become a center of excellence for AI experimentation. An AI pilot or proof of concept (POC) is “your chance to make sure a tool will actually deliver results for your organization: improving developer experience, accelerating development, and ensuring your engineering practices still align with your foundational definition of excellence,” explained [DX CTO Laura Tacho](#).

Tacho offered seven decisions that she recommends to guide your AI tool evaluation:

1. **Goals:** Know exactly what you are trying to achieve, like finding a new use case or deciding what teams get more budget.
2. **Tools:** Pick the scope, including whether you are trialing one tool or comparing several tools.
3. **Cohort:** Likely the most important decision, she contends: Are you testing impact on individual or organizational productivity?
4. **Duration:** Set a start and end date. As Tacho said, “too short and you miss the real picture; too long and you lose momentum.”
5. **Metrics:** “Measure impact the same way you’ll measure it at scale,” she said.
6. **Scoring:** Don’t forget other metrics, like procurement alignment and security certifications.
7. **Decision ownership:** Before the experiment kicks off, determine who decides if it’s the right tool for the outlined team and use case.

Don’t forget to clarify where and how teams are allowed to run these experiments. A lot of enterprises are choosing to run them in on-premises sandboxed environments for a couple months to validate the AI use case before considering rolling them out to the full team, division, or organization.

Finally, make sure you have clear channels of communication — perhaps biweekly demo days or lunch and learns — where employees feel safe to share AI wins and, especially, when AI doesn’t work out.

CHAPTER 04

Data Security Backed by Engineering Best Practices

With great speed comes increased responsibility. AI-driven productivity cannot come at the cost of data security, reliability, or privacy. And, while AI safety must be a priority, it can only exist alongside engineering best practices that have become more urgent than ever.

New Risks Enterprises Face With AI

As organizations accelerate their AI adoption, they must reckon with both new and existing threats.

Enterprises need to address new risks across the AI life cycle, including:

- **Governance and testing gaps:** There are already holes in policy enforcement and testing coverage at most organizations. How can the chief information security officer's (CISO's) office keep up with an exponential increase in code and content creation? How can security teams maintain a view across a much larger codebase? Do you have enough functional and nonfunctional tests to adequately cover this much code before it's released?
- **Holes in the CI/CD pipeline:** When things inevitably go wrong, do you have the progressive delivery best practices in place to reduce the blast radius? Do you have automatic rollbacks in place? As AI quickly moves

outside machine learning departments, how do you manage and isolate access across your life cycle? Do you have DevSecOps down?

- **Complex supply chain:** As data science inevitably folds into your engineering organization, securing your software supply chain will only get harder. Have you documented all the components in your AI life cycle so that you can prevent and respond to vulnerability risks?
- **Bias in data and AI:** As AI agents begin acting autonomously, how will you ensure models operate as intended and without bias? How do you communicate that trust internally and externally?
- **AI-specific risks:** Are your technical and nontechnical teams ready to protect against malicious actors using modern techniques including prompt injection, data poisoning, data leakage, and inference attacks?

Data Security Is a Top-Level Priority

To manage these new risks, your AI strategy must include a multilayered, platform-led approach to AI security at scale.

An organization's differentiator is its data. The potential for risk to this critical asset may be what makes an enterprise reluctant to fully embrace AI.

Raghu Moorthy, global director and principal engineer at Intel, outlined the C-suite's top data security concerns:

- Protecting sensitive data in memory or RAM.
- Regulatory and compliance requirements, which mandate that sensitive workloads remain confidential even from cloud providers.
- Insider threats and malicious actors that have access to view or manipulate data.
- Slowness to modernize, in part due to lack of trust in third-party cloud providers, with concerns about legal access requests and unauthorized access.
- Multitenant and shared infrastructure risks in the public cloud.

“AI-driven productivity cannot come at the cost of data security, reliability, or privacy.”

Due to these considerations, plus the sheer volume of modern-day attack vectors, organizations are intentionally moving away from traditional perimeter-based security models that rely on “assumed trust” within firewalls.

In its place, the [zero trust security model](#) has become the gold standard. This model of “deny all, verify all” is grounded in authentication and authorization via gated access, depending on several factors:

- Who is on each side of the gate?
- What data is allowed to move through the gate?
- How is the gate protected?

Zero trust stands atop three pillars of data security:

- Data in transit: Making sure that data is being securely transmitted from one point to another.
- Data at rest: Encrypting and controlling access to databases.

- Data in use: When data is being used, for example, on a computer.

While the first two use cases are already commonly addressed at the enterprise level, Moorthy said that the latter data use case is still underaddressed.

“You spent a lot of time training that model. You want to put that model in a place, and you don’t want someone else to steal that model,” he explained.

Confidential computing is an emerging trend to address this risk. Moorthy explained confidential computing:

“You take a model, and then you put the model in the memory, and then you scramble it up. So even if you get access to that hardware and to that memory, you wouldn’t know what’s going on.”

An AI-ready platform should enable confidential computing capabilities across your whole AI stack. This will help protect your data in cases such as when a laptop is lost or stolen.

Continuous Delivery Best Practices

The most adopted AI use case right now is code generation. [Microsoft](#) and [Google](#) have reported as much as 30% of their code is written by AI. That pure increase in lines of code has a knockdown effect on every aspect of your business, from test coverage to security, from code reviews to marketing and customer success when rolling out new features. Each of these teams already had a backlog to reckon with, and they're now handling more features being shipped faster at potentially lower quality.

A big thing to keep in mind is that most of these large language models (LLMs) have been trained on the internet, so the code they generate could contain errors, hallucinations, or unpatched vulnerabilities.

"There's this massive focus on productivity and creating code as fast as possible. Backlogs are endless, and everybody's complaining that the IT or the tech department's not fast enough, and we need to get more features out," Rachel Laycock said. "The internet isn't

necessarily full of great code, and generating more of that is not necessarily good for us.

"Fact: AI-generated code does not equal better code. AI-generated code currently takes longer to debug, and [76% of developers said they need to refactor it](#).

If organizations didn't have CI/CD, continuous testing and documentation, and security under control before AI, how can they possibly keep up as code volume increases? Adding more code to the fire will only burn more holes in your delivery rate. Indeed, the [2024 DORA report](#) found that increasing AI adoption reduced delivery stability by 7.2% and delivery throughput by 1.5%.

DevOps Best Practices

The most important tools for AI may not be AI-based. Rather, they are the tools that support strong DevOps, CI/CD, and automation best practices.

“The teams that were doing well with non-AI software delivery at velocity will continue to do better than those that were still patchy in their automation and behind in adopting modern engineering practices, like A/B testing, blue-green deployments, automated testing, CI/CD,” Hannah Foxwell explained.

Continuous, automated testing must be a requirement to move to production, especially for AI-generated code. AI can help by speeding the creation of unit tests and running them on newly generated code.

Even if you are releasing AI-generated code to production, you don’t have to release it to all users at the same time. Progressive delivery, which RedMonk co-founder James Governor defined as “continuous delivery with fine-grained control over the blast radius,” is as important a practice as ever.

Progressive delivery routes traffic to a specific subset of users before deploying it more broadly so that testing in production doesn’t have to be a massive risk. Methods include:

- Canary testing
- Blue-green deployments
- A/B testing
- Feature toggling
- Service meshing

Apply AIOps practices — automation that enables AI capabilities and tools for operations — across your software delivery life cycle. To achieve automated remediation, integrate observability tools with [event-driven automation](#) and generative AI — like feature flags and automated rollbacks — so systems can detect events, propose responses, revert to previous state, and execute fixes with minimal human intervention.

There are potential reliability impacts with “more changes being shipped to your users more frequently — unless you have good recovery modes built into your products and your platform,” Foxwell said.

Monitoring and observability must also be in place to understand the user experience, so you can alert the right people about issues worthy of investigation (such as a drop off in user rates). These are also critical for gauging the immediate and long-term impact of AI-generated code, answering questions like: Are you increasing speed to production at the expense of maintainability?

AIOps goes beyond traditional observability by also monitoring, predicting, and automating actions related to the high-performance hardware running AI models. A platform backed with AIOps acts as an intelligent control layer that tracks:

- GPU utilization and efficiency metrics

- Memory capacity and bottlenecks
- Hardware health and any errors

This enables the platform to automatically reroute workloads to perform more efficiently and cost effectively.

“The teams that were doing well with non-AI software delivery at velocity will continue to do better than those that were still patchy in their automation and behind in adopting modern engineering practices, like A/B testing, blue-green deployments, automated testing, CI/CD.
— Hannah Foxwell

CHAPTER 05

Future-Proof Your AI Strategy

It seems like every week we have a new winner of the AI GOAT (greatest of all time) contest. The reality is that things are changing so fast that no organization can make long-term bets on any AI tooling.

That doesn't mean all long-term AI strategy and investment is useless. You should be investing in training for your employees. You should be looking to move more services, workloads, and applications to containers, microservices, and the cloud. You should be further adopting DevOps best practices, expanding your testing coverage, and increasing your automation and CI/CD. You should be unifying, cleaning, and categorizing your data. And you should be ensuring compliance and security are major considerations in every step.

Beyond these must-haves, flexibility is key to AI success. This is best served with an open source ecosystem, strong infrastructure, and platform engineering.

Open Source Platform Pluggability and Agility

Vendor lock-in stifles the ability to respond to rapid change, while finding the right set of best-in-class open source AI tools supports flexibility. Back this with a platform engineering strategy that allows you to plug in tools, rolling them out to the whole organization once vetted.

“Vendor lock-in is a huge concern. Just as orgs moved away from monolithic CRMs and ERPs, they now need modular, composable architectures that let them swap components and test innovations quickly,” said Dana Lawson, CTO at Netlify. “Security is the constant: Enterprises must put guardrails, proxies, and inference controls in place so that data entering and leaving models is governed responsibly. AI may be new, but the enterprise responsibility to protect data and maintain trust hasn’t changed.”

The right platform delivers these essentials with breadth and depth that give everyone visibility into the open source supply chain, architecture, infrastructure, cloud, and hardware,

with the right levels of abstraction for ease of use.

Platform engineering involves laying down “golden paths” or the simplest, preferred way to do repeating tasks and processes. It often centralizes processes and documentation in a searchable manner. A platform is the simplest way to support service and data discovery — including documenting what’s running through third-party APIs and apps — which, in turn, enables innovation and reusability.

The internal developer platform (IDP) has become the preferred way to continuously serve IT stakeholders. It sets up users for self-service, often with a search or chat interface. It makes onboarding faster by managing role-based access control (RBAC) and access to different tools and data sources.

A platform allows developers and other tech workers to build on top of your standards and infrastructure. This gives them the freedom to build without

Platform Engineering for Data Science

threatening your security, compliance, and reputation. Plus, as your cloud computing bills inevitably skyrocket, sharing the architecture and underlying infrastructure will cut maintenance costs and shared compute.

As achieving the full potential of a cross-organizational strategy hinges on secure access to data, the effectiveness of that data relies on having no silos. Platforms centralize data in a secure and gated way. This centralization then enables AI, when directed, to gain a more holistic view of the whole software development life cycle, which in turn makes it all more measurable.

In this time of rapid change, a platform creates a consistent user experience, helps break down silos, and opens opportunities for learning from peers. These could make the platform the best way to center data science within your IT organization and overall AI strategy.

After all, RBAC, security, and onboarding use cases are the same between data science and IT. Beyond that, platform engineering can help meet data scientists' compute requirements, Wesley remarked.

Data scientists have been using their personal computers with GPUs to do AI work. But when they needed more GPU resources, they couldn't make use of other colleagues' computers and aggregate those resources to theirs. But with a platform based on Infrastructure as Code (IaC) and automation, teams can schedule on infrastructure nodes that have GPUs for more efficient resource sharing.

The right platform addresses model drift too.

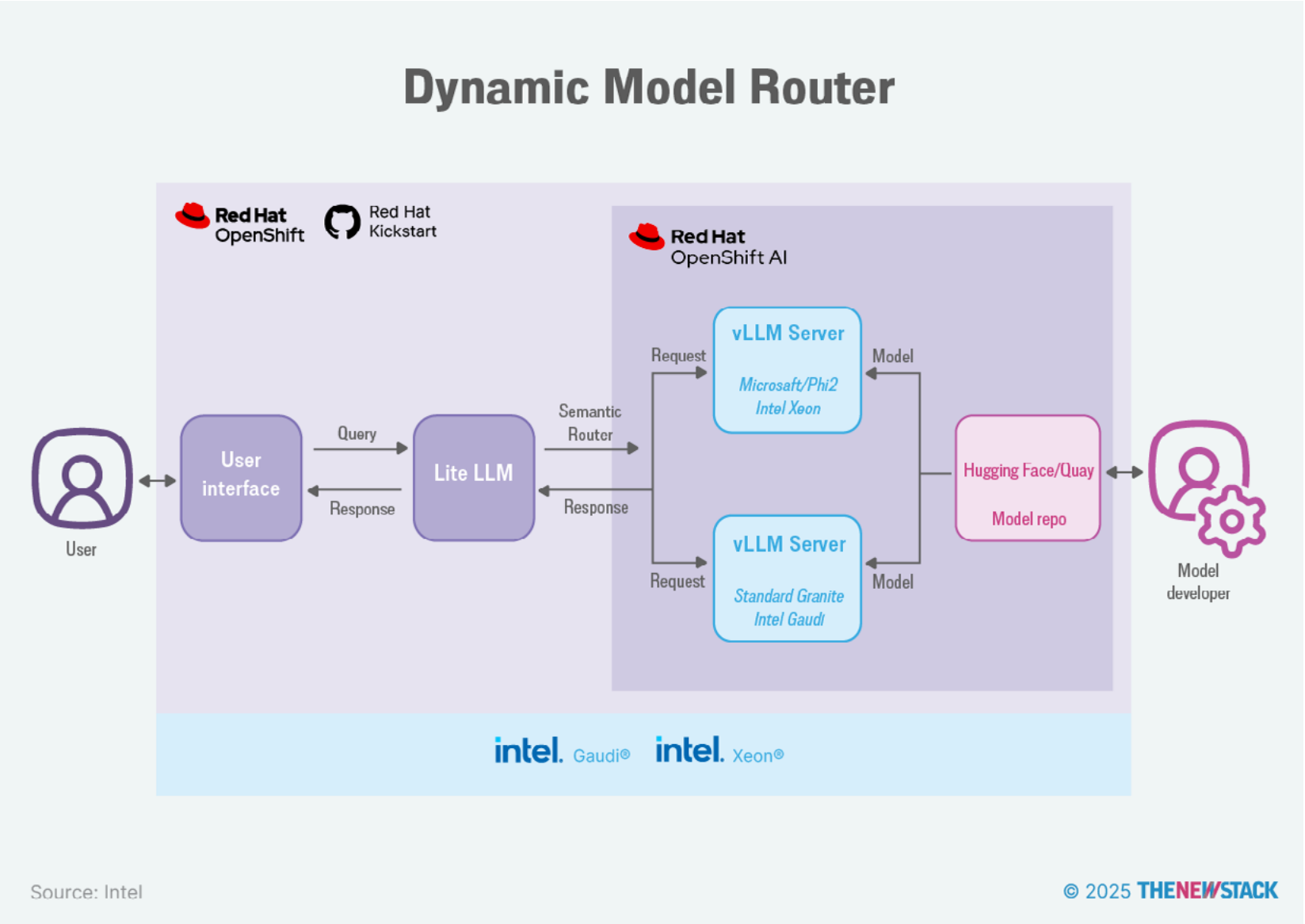


FIG 5.1 A user’s query is evaluated to be simple or complex, and a semantic router routes that request to Xeon for simple queries and Gaudi for complex queries. Each model deployed on Xeon and Gaudi in OpenShift AI uses serving runtimes with vLLM for optimized throughput and memory. This dynamic model router solution assists with total cost of ownership of hardware platforms.

“When your SLOs aren’t being met, you have to go through that process to take out what’s out of sync and retrain the model,” said Joe Giordano, chief architect in the Red Hat field CTO organization. With the right platform, when models drift and lose accuracy, you should be able to automatically pull in data, retrain, and then redeploy the model with strong observability in place throughout.

Most enterprises are already running machine learning workloads across PCs, on the edge, in data centers, on premises, and in the cloud. Many of these workloads were placed based on the convenience of the engineering team that built them. Now, with platform-led AI, you can more easily, automatically leverage the best compute per workload.”

A typical AI pipeline has, say, 50 tasks because I have to do a series of things to get to my end goal. There are different kinds of models in the process — some bigger, some smaller — because it’s a chain of things that you have to do,” Moorthy said. “A large number of tasks in that chain don’t need heavy-duty compute” like GPUs. CPUs are sufficient for many AI tasks.

By using a platform overlay, which understands and then can automatically

route different workloads across different nodes, clusters, and compute providers without breaking that chain, he said, “the overall cost of your AI pipeline is now lower because you’ve replaced the more expensive parts with [less expensive] compute capacity.”

Then, by focusing your AI strategy on open source and open ecosystems, you are not only able to more quickly respond to this rapidly evolving market.

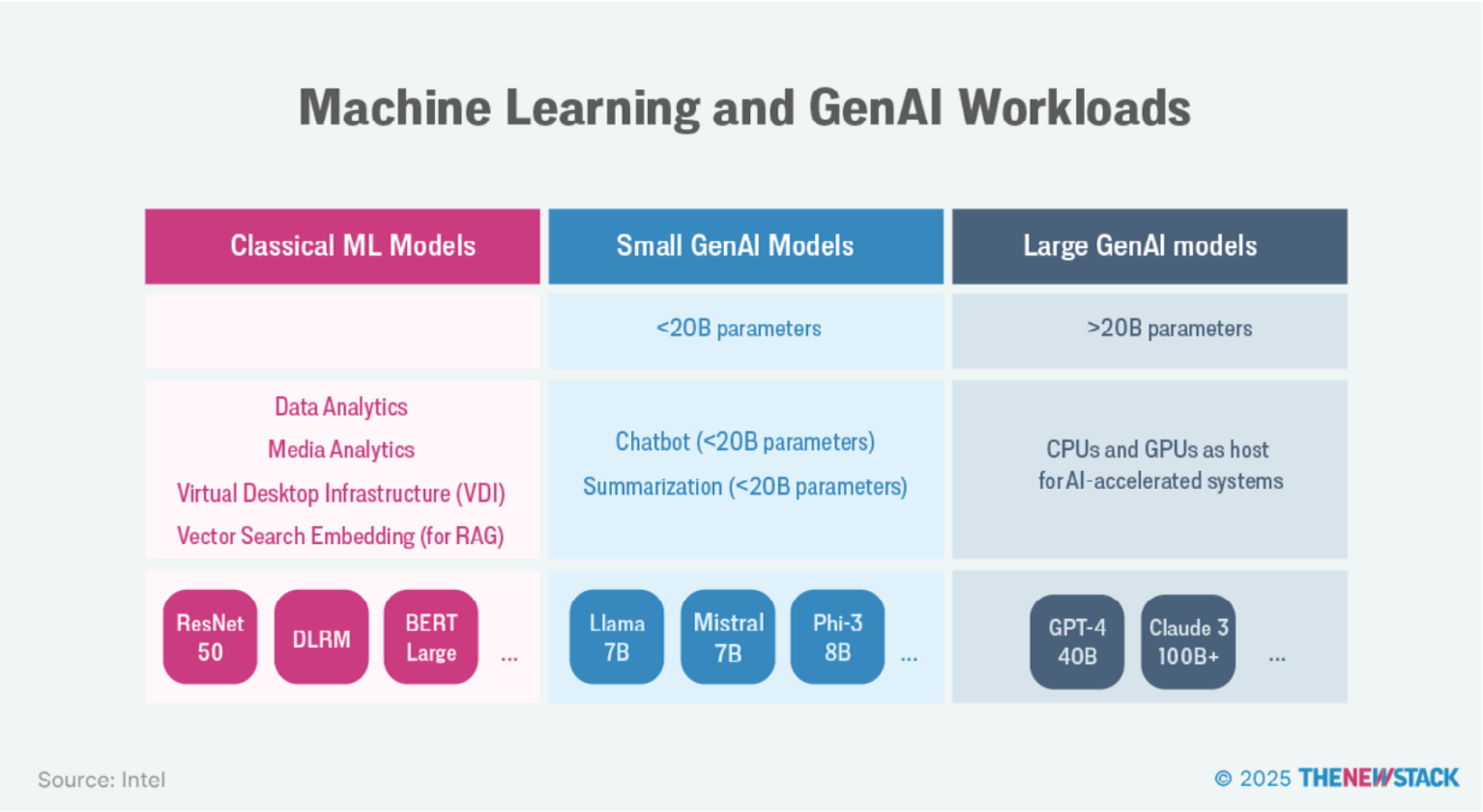


FIG 5.2 Classical ML models are best used for data and media analytics, VDI, and RAG, whereas small GenAI models are appropriate for chatbots and summarization, and large GenAI models are used for AI-accelerated systems.

Prepare for the Agentic Future

You can also focus on getting the best total cost of ownership, Moorthy continued, with the right hardware for the right purpose — which also leads to a lower carbon footprint.

Platform engineering brings different tools, data sources, and teams together. For this, a consolidated platform strategy is considered one of the best ways to [measure the ROI](#) of your AI strategy. This is because it facilitates a “single pane of glass” view for business and technical stakeholders, while still maintaining auditable data sovereignty.

And when you eventually find the proven use cases that can work across different business and administrative roles, a platform might emerge as the best way to do a cross-organizational rollout.

For most enterprises, generative AI is the present. But preparing the data, processes, infrastructure, and more for the AI of now sets you up for success with the agentic AI of the future.

Agentic AI is the practice of deploying AI agents to achieve goals with minimum human intervention. An AI agent is an autonomous or semiautonomous system that can make decisions and take actions. It’s like a software developer, if each developer only did one task and only had one responsibility.

Software developers are complex, multitasking, problem-solving creative workers. That means these humans are getting good at orchestrating a mix of humans and AI agents to deliver software faster — and eventually a team of agents working alone. Similarly, it’s predicted that other, less technical, domain experts across the business [will soon be able to direct swarms of AI agents](#) ready to perform repeating, noncreative tasks.

An early AI-plus-platform use case includes templates for agentic AI workflows. Once you’ve proven AI can repeatedly do a task (often menial) reliably, then you can allow an AI agent or swarm of AI agents to carry out that task with the right guardrails and testing in place.

“Enterprises face a critical challenge as AI enters their workflows: adapting CI/CD pipelines without ripping out the systems they’ve spent years fine-tuning. Most can’t ‘rip and replace,’ so they need to modernize piece by piece, with composable architectures that integrate AI agents into existing processes,” Netlify’s Lawson said.

“This is the shift from developer experience (DX) to agent experience (AX) — not just how a developer interacts with software, but how AI agents themselves interact with enterprise systems.”

LinkedIn’s tech stack enables engineers to build AI agents that can “think, plan, and act in collaboration with users,”

[wrote LinkedIn distinguished engineer Karthik Ramgopal](#). Lessons learned from LinkedIn’s deployment include:

- Reuse existing infrastructure and provide strong developer abstractions in order to scale complex AI systems efficiently.
- Design for human-in-the-loop control while enabling agents to operate autonomously when appropriate.
- Recognize that observability and context engineering are necessary for debugging, continuous improvement, and delivering adaptive, personalized experiences.
- Adopt open protocols to enable interoperability and avoid fragmentation as agent ecosystems grow.

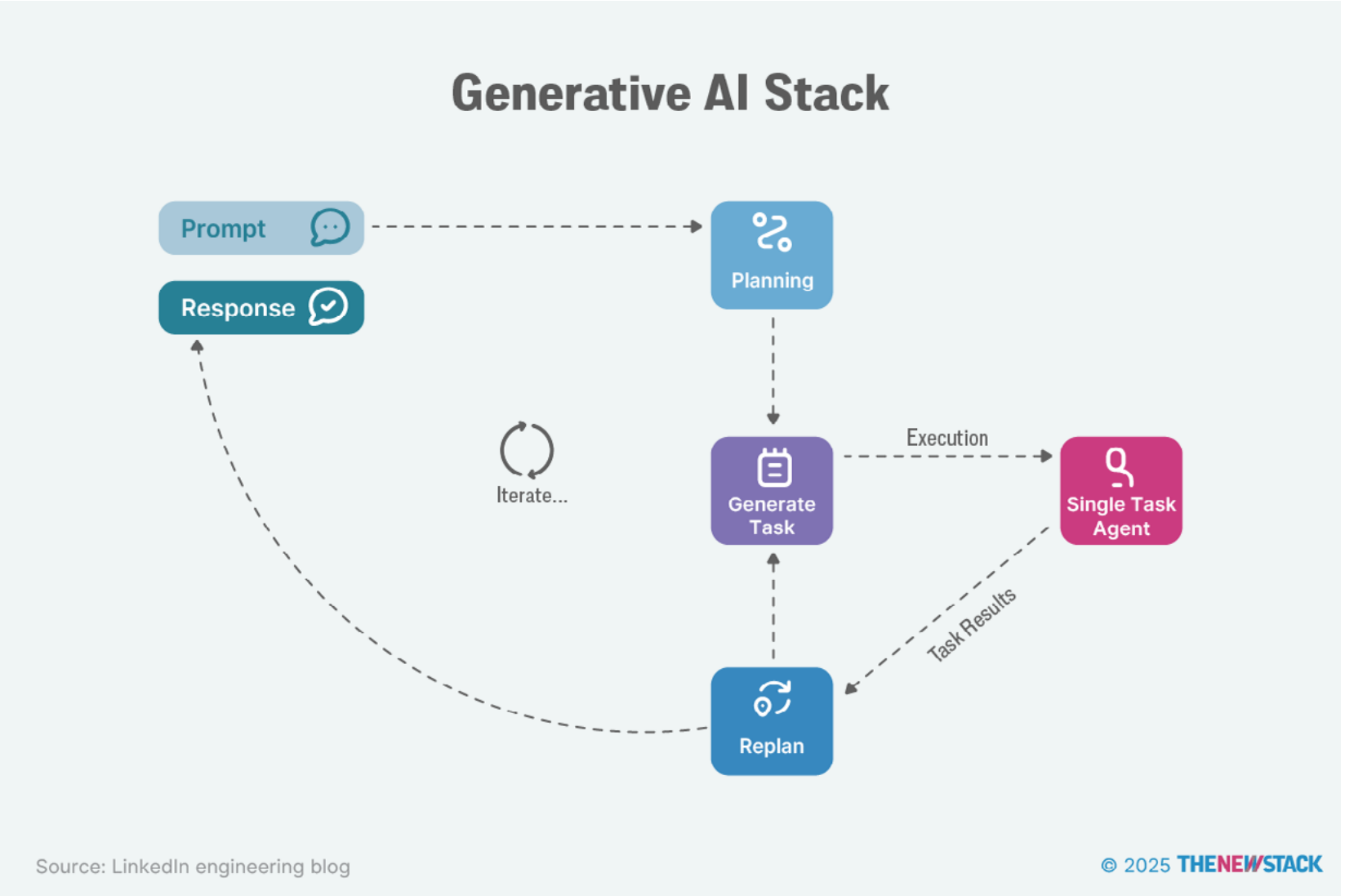


FIG 5.3 LinkedIn’s GenAI architecture.

“Unlike traditional software, GenAI agents don’t just follow hard-coded logic; they use LLMs to interpret goals, make decisions, adapt to changing conditions, and even break tasks into subtasks before solving them,” explained Ramgopal. “Given the cognitive limitations of current LLMs and to ensure that human operators have control, agents need to operate with a ‘human in the loop’ approach.”

LinkedIn has agents seeking clarification, feedback, and approvals from humans at key decision points. For the foreseeable future, this will be how

AI agents should be treated: with care and human control.

No one I spoke with has seen agentic AI workflows being set up and run by nontechnical stakeholders.

“One of the biggest misconceptions I have felt, having spent most of this year building agents, is that [building agents] is something for your average business users,” Foxwell said. “As soon as agents start to do anything of any reasonable complexity, it becomes an engineering-shaped problem.”

Prompt engineering, a hot topic of 2023 and 2024, has mostly faded away. This year and next, she said, will be all about [context-based engineering](#), which is designing, organizing, and managing the information ecosystem surrounding an LLM or AI agent.

Foxwell said every time she gives agents a list of more than 10 things to do, “the context gets exceeded, and they go completely awry. The more tools you give these agents, the more easily confused they get.”

Minimizing context would certainly leave the agents less confused, but more context means the results should be more accurate and consider your organizational guardrails, codebase, and history. When current AI agents can handle simple tasks, she said, it’s probably better to complete those tasks in another, less resource intensive way, like automation.

According to software engineer and technical lead Peter Szel, the best application of early AI agents is when a

task or series of tasks follows a pattern. In his experience, about 80% of tests follow the same pattern, so you can scaffold a testing framework and train an AI agent to run that test, he said. Similarly, a lot of endpoints follow the same pattern, so, “if you show the pattern to the AI, they’re going to do a great job with that.”

“This is the shift from developer experience (DX) to agent experience (AX) — not just how a developer interacts with software, but how AI agents themselves interact with enterprise systems.

How Much Should AI Really Be in Charge?

On the other hand, he warned, sometimes, if the task is very simple, it takes more time to craft the correct prompt than for an engineer to execute it. However, mimicking the “Rule of Three” for automation, if it’s something your team has to do the exact same way at least three times, then it could be a good task for an AI agent.

On the other hand, “if the task is too complex or the bug is too hard to find, I see AI struggling,” Szel said.

Good use cases for agentic AI, he said, are in the middle ground: “If it’s medium complexity. There is a pattern I can show. There is something similar implemented in the codebase. Nothing too performance intensive. Nothing security critical.”

And, he emphasized, AI agents aren’t ready to execute anything without a human reviewer in the loop.

This may change dramatically in the next year or two, but, for now, your enterprise must decide — at the organization level, at the use case level, and at the team level — how much autonomy you want to give AI. Generative AI still leaves a lot of control with the user — except maybe in public-facing chatbots — but, as you move into coding agents or even swarms of agentic AI, this changes.

“The two extremes are: Let everything be done by AI, [or] let everything be done by hand,” Szel said. “But there are multiple levels where, in the middle, for example, I define the architecture, I define the patterns and practices I want to follow, and I create the plan,” while AI acts on these things.

Szel has identified [five levels of AI delegation](#) that aim to help enterprises get past the binary of “AI versus no AI”:

- **Level 1 – Manual coding:**

Especially for critical systems or when an engineer is learning something new.

- **Level 2 – Target assistance:** Ask AI for single functions or classes.
- **Level 3 – Step-by-step collaboration:** An engineer creates the plan, AI reviews and iterates on it, and then the AI asks for approval for each task.
- **Level 4 – Plan then execute:** The engineer defines the problem, then AI creates a plan. The engineer then refines it and the AI executes it autonomously.
- **Level 5 – Vibe coding:** The engineer writes prompts in natural language, and the AI writes and executes the code without review. This works best for brainstorming and rapid prototyping.

The first two levels aren't as fast, he remarked, but your team retains architectural oversight. The last two levels definitely move faster, but even the engineer is unlikely to know the implementation details, which makes it much harder to find issues.

Szel contends that Level 3 is the “sweet spot,” where the developer retains architectural oversight but delegates the grunt work.

Everything also depends on the engineer's experience. A junior to midlevel developer, who must optimize for learning, should avoid Level 5 and focus on Levels 1 and 2. It's more important to get in the habit of “harshly critiquing” the AI-generated code:

1. Try to come up with why it would fail.
2. Think about similar or alternative approaches.
3. Consider how this code fits into the overall application architecture.

Level 4 is different from [specification-driven AI development](#), where the spec is the single source of truth and the AI agent executes. Szel argues that spec-driven development takes away the creative work of engineering.

“The idea of working just on specification and letting the agent generate the code sounds horrible to me,” he said. “Even though it might work, I don’t really want to work on specifications eight hours a day.”

Additionally, spec-driven development, he argues, will require developers to maintain not just more code, but more specifications too.

Be sure that your engineers and data scientists curate the context window for your developers. This includes the codebase or code files you are working with, along with the coding and architectural guidelines — anything you need the LLM to be aware of when you assign it a task, Szel said.

If engineers leave the context window empty, he warned, the LLM will surely hallucinate. But, if the context window is too cluttered, the LLM will have difficulty figuring out what to do.

Then, he continued, ask the coding agent to test its results by trying to run the code itself.

Szel “asks the agent to start with creating the automated tests first, and that’s something I review before they even get started.”

This offers several advantages, he said, because “if they can write a correct test, it means that they at least understand the problem they are solving, and then they can go ahead and implement. Then, after they’ve implemented the new feature or they fix the bug, they can run the test themselves and check whether they actually fixed the bug or implemented the feature.”

Measuring the ROI on Your AI

Science without a plan of action and measurement is no longer science. And yet enterprises have spent the last couple years writing blank checks to AI adoption.

Just like every organization is different, so are every organization's goals. Which means metrics can't be one-size-fits-all.

According to DX's Tacho, every organization must:

1. Define engineering performance for their organization.
2. Determine the specific metrics to measure the impact of AI on that performance.
3. Then figure out what to measure that supports these goals.

So much of AI adoption has been focused on code generation and its opportunity to speed time to delivery. But throughput cannot be your only metric, because speed to release can come at the expense of quality, security, and/or reliability.

DORA Metrics

Certainly the DORA metrics are a good place for any enterprise to start. These include [four key metrics](#) that should be monitored before, during, and after any AI change:

- Change lead time
- Deployment frequency
- Change failure rate
- Failed deployment recovery rate (formerly mean time to recovery or MTTR)

Developer Surveys

Beyond these industry standards, it's a good idea to check in with all employees, perhaps weekly given the pace of change, to see how their AI adoption is going. Ask specific and quick questions about developer satisfaction with tools being tested like:

From 1 to 10, with 1 being significantly worse, 5 meaning no change, and 10 meaning you couldn't live without it, how do you feel the adoption of [AI tool] has affected your productivity?

While it's faster and simpler to collect quantitative metrics, [don't forget qualitative feedback too](#), recommends Adnan Masood, author of "Responsive AI in the Enterprise." This data comes from regular developer surveys, engagement scores, and sentiment analyses.

"I've found that developer happiness and satisfaction often directly correlate with productivity, retention, and overall quality of work," he wrote.

Cost Considerations

Intel's Moorthy emphasized that another core AI metric to consider is the ongoing cost of AI workloads.

By using efficient, automated resource rerouting, you can combine GPUs and CPUs to lower the overall total cost of ownership. The right monitoring tool or platform will flag whether a model is spiking its usage, making it a target for investigation and fine-tuning.

Another cost — to business and reputation — is any data leakage. Enterprise AI requires continuous auditing and scanning, alongside respective metrics when this inevitably happens, including:

- Mean time to detect
- Mean time to recover, respond, or contain
- Volume or percentage of data leaked
- Overall cost to business

This strategy must also include who you're going to communicate these numbers to inside and outside the organization.

Don't Forget the Broader Context

Also make sure metrics have context. Take the aforementioned Harness study that found that 68% of developers spent more time resolving security vulnerabilities in AI-generated code.

Is that because AI-generated code is inherently less secure? Maybe. But don't negate the possibility that developers are spending less time on toil and more on value-driven work like patching vulnerabilities faster. Or perhaps AI-automated testing is reducing the time to detect incidents. You don't know until you measure regularly.

"Research on technology adoption shows that success isn't driven by how quickly or widely a tool is rolled out — it's driven by how well it aligns with real, high-impact problems," [said Lizzie Matusov, CEO of Quotient](#). "The teams with the most success are the ones starting with their pain points — long cycle times, unstable requirements, too

much work in progress — and asking: 'Could AI meaningfully move the needle here?'"

In other words, she continued, part of success with AI is about rightsizing AI to fit within your organization.

Technical Debt

Another important AI concern is rework. LLMs often give the answer that is most likely to be accepted, without considering the whole codebase or even if it's the best or most accurate response. While AI can be a powerful productivity booster, this exponential increase in code is also increasing technical debt.

"When the use of AI-generated code is scaled rapidly or applied to brownfield (legacy) environments, the risks are much greater and much harder to manage," wrote Edward Anderson, Geoffrey Parker, and Burcu Tan in [MIT Sloan Management Review](#).

[Key metrics for rework and quality](#), according to software consultancy Waltham, include:

- **Static code analysis**, which checks for poor design, complexity, repeat code, and bug density.
- **Functional correctness and security**, including testing and scoring for compliance, security vulnerabilities, and whether it even works.

Code churn — the rate at which lines of code are changed or deleted shortly after being written — is another quality flag.

“Time to find bugs” is a good signal that either the code quality is decreasing or that your developers are too distant from the generated code. And any rise in security vulnerabilities should trigger an investigation of the AI tooling and the processes that could’ve led to it.

Most notably, research in the journal “Empirical Software Engineering” found

that [AI coding tools miss 60% of security vulnerabilities](#) — so you need to measure your testing automation to make sure it doesn’t miss them too.

Developer Productivity

The time it takes someone to feel successful in a new company or on a new team is crucial to long-term productivity and tenure. An interesting measurement is time to onboard, which, in engineering, is most commonly quantified as time to the 10th pull request.

“Onboarding is an expensive and time-sensitive phase of the developer journey, both for new hires and developers switching projects,” Tacho said. “For large enterprises, it can take months before new hires are fully productive, creating both financial costs and delivery risks.”

Considering the average salary per engineer, this is an important measurement. Tacho recently analyzed data from six multinational enterprises

and [measured how quickly new hires reached their 10th PR](#). With daily AI usage, engineers hit their 10th PR in 49 days; without regular AI usage, it was 91 days.

AI usage is also dramatically closing the knowledge and experience gap between new hires and those who’ve worked at an enterprise for more than a year, [DX research found](#). Developers with longer tenure traditionally have a somewhat higher PR throughput than recently hired peers. But, when AI is “meaningfully embedded into daily

team and individual workflows,” Tacho found, AI acts as a force multiplier.

AI Measurement Framework

While each company is different, all need a standard way to measure the effect of AI over time.

As a place to start, DX has released an [AI Measurement Framework](#), which combines different frameworks to create a set of recommended metrics across three dimensions.

AI utilization metrics	AI impact metrics	AI cost metrics:
• AI tool usage at the daily and weekly active user levels, at team levels, and different job roles • Percentage of pull requests that are AI-assisted • Percentage of code committed that is AI-generated • Number or percentage of tasks assigned to AI agents	• AI-driven time savings (dev hours per week) • Developer satisfaction • Pull request throughput • Perceived rate of delivery • Code maintainability • Change confidence • Change failure rate	• AI spend, both overall and per developer • Net time gained per developer (time savings minus AI spend) • AI hourly rate (human-equivalent hours divided by AI spend)

Most In-Demand Skills in the Age of AI

As we enter an “age of robots,” are the most human qualities still the most needed?

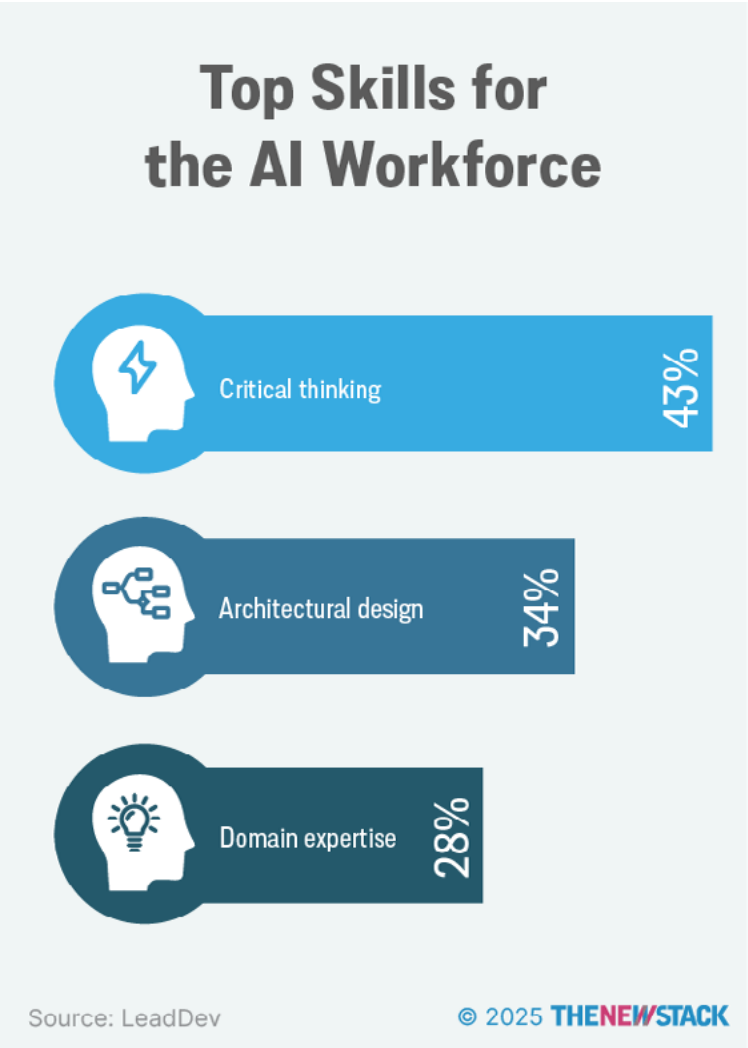


FIG 5.4 Engineering leaders say critical thinking is the most important competency for software engineers.

According to LeadDev’s “[AI Impact Report 2025](#),” the most in-demand competencies are critical thinking (43% of respondents) and architectural

design (34%), while 28% said they require more domain expertise on their development teams. Considering the importance of context to AI, especially in customer service chatbots, the latter need will only increase.

AI should be seen as a human-augmentation tool, which means these humans need to be onboarded and trained in how to work with this new AI partner — so they can make sure they are training the AI right back.

As generative AI tools take on more of the code writing, new data from engineering optimization system Uplevel shows that two-thirds of engineering leaders believe [quality assurance \(QA\) and validating AI outputs are the most critical skills](#) for their teams. About a third also emphasized the need for both performance monitoring and optimization, as well as system architecture and integration skills.

Junior engineers need editing skills — the ability to read a block of code that they didn’t write, understand what

works and what doesn't, and then communicate that back to the AI for iterative improvement. Everyone needs to get into the habit of prompting the AI to define the security flaws within code and to consider the impact of code within the larger codebase.

While a modern platform engineering approach automates and abstracts out a lot of AI's complexity, as data science and engineering move closer together, both sides will need knowledge about MLOps. MLOps applies DevOps principles to machine learning (ML) systems, with an important focus on cross-team collaboration, automation, continuous delivery, reliability, and scalability.

Therefore, Omotayo Adeleke, backend engineer at DoWell Research, recommends everyone working in tech acquire skills in:

- **Prompt engineering:** Creating effective instructions for AI models.
- **Model tuning:** Optimizing model parameters for specific use cases.
- **Telemetry literacy:** Reading and understanding model metrics and behavior.

Finally, every employee needs to know how to flip the switch from using generative AI chatbots as one-way search engines to having conversations that lead to better results, all with a constant emphasis on data quality, privacy, and protection.

CHAPTER 06

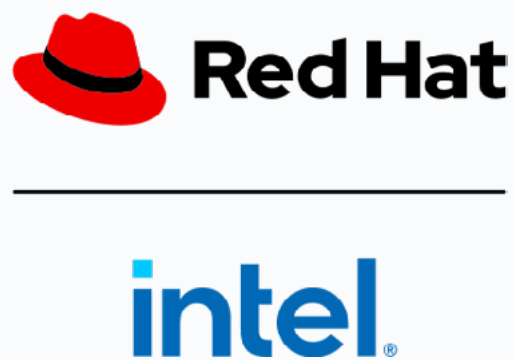
Embrace the Opportunity of AI

AI may feel ubiquitous right now, but, in a rush to adopt, many organizations are putting themselves and their reputations at risk. Currently, most of the benefits of AI remain in those smaller, proven improvements that shouldn't be ignored — they have notable impact at the individual level. These benefits really add up across hundreds to hundreds of thousands of employees.

Then, by adopting an intentional, measurable cross-organizational AI strategy, you will be able to unlock the true productivity gains and solve those previously impossible problems.

Just don't forget to slow down enough to properly onboard. You may end up dogfooding an internal use case that can benefit your customers too. Delta Air Lines rolled out [AI-based training and career development](#) for employees a year before delivering any AI improvements to customer experience, as it bet that improving employee experience would increase the latter.

AI is an opportunity to take the boring of our day to day and to help everyone do what they do best. AI can be the thing that emphasizes each organization's differentiation — but don't believe that AI is that differentiator. Ground your AI enterprise strategy in your people, and then your processes and technology, to get the best out of an AI-enriched future. Good luck!



SUMMARY

Our goal is to deliver innovative, security-focused solutions that help our customers gain a competitive advantage. Red Hat and Intel innovate together to deliver advanced software-defined infrastructure that offers data center agility, scale, and flexibility. Our long partnership goes back more than 25 years, and today we provide cloud-ready network architectures based on industry-standard platforms that are suitable for high-performance needs.

KEY FEATURES

- Intel works closely with Red Hat to implement the newest technologies and software with high degrees of optimization. Customers gain workload-optimized containers and integrated solutions for enterprise grade cloud architectures.
- Powered by Intel's Device Plugins Operator for Red Hat OpenShift Container Platform, Intel Software Guard Extensions (Intel SGX) provides advanced data protection.
- Using Intel AI Frameworks and Tools with the Intel Distribution of the OpenVINO Toolkit and Gaudi Software, developers can learn, test, and deploy AI using Red Hat OpenShift AI.

About the Author



[Jennifer Riggins](#) is a tech storyteller and journalist, event and panel host. She bridges the gap between business, culture, and technology, with her work grounded in the developer experience. She has been a working writer since 2003, and is based in London.

Disclosure

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